Ryan Strawbridge

508.594.9661 | strawbridge.r@northeastern.edu | LinkedIn | Portfolio

Education

Northeastern University – *Boston, MA* May 2027
B.S. in Mechanical Engineering (BSME) *GPA: 3.58*
M.S. in Mechanical Engineering (MSME) Mechanics & Design Concentration, Expected Dec. 2027, ***Available to work in July 2027***

Coursework: Continuum Mechanics | Heat Transfer | Fluid Mechanics | Solid Mechanics | Thermodynamics | Material Science

Activities: AeroNU Propulsion Member | Running Club President | Rhythm Guitarist in Alt. Rock Band

Achievements: DoD Security Clearance | CSWA Certified | Lean Six Sigma Certification | 800th in the Boston Marathon

Professional Experience

Commonwealth Fusion Systems Devens, MA
Manufacturing / Equipment Engineering Co-op (Term 2) *Jan 2026 – Jun 2026*

- Designed and validated a 2,100-lb hand-portable lifting frame to **BTH-1** and **ASME B30.20** using statics, buckling, threaded-joint, and weld analysis; earned structural engineering and safety approval and **reduced lift time by 30–45 minutes**
- Created a modular CNC seam-cutting fixture using **GD&T** and dowel-pin datums for repeatable alignment; enabled early qualification testing and **reduced projected operation time by over 80%**
- Deployed a low-cost adjustable fixture using welded tabs and off-the-shelf linear rails and carriages to maintain cable alignment during insulation wrapping, improving process robustness and **reducing labor by 0.5–3 hours per magnet layer**

Manufacturing Engineering Co-op (Term 1) *Jan 2025 – Jun 2025*

- Re-designed a hydraulic piston-driven bending machine in **Siemens NX** using bolt safety analysis and first-principles calculations, cutting operation time and reducing injury risk; **still in use over a year later, saving hundreds of labor hours**
- Developed a predictive equation relating bend offset, bend radius, and arc length through iterative spring-back experiments and data interpolation, **eliminating trial-and-error die design** and enabling future cable-bending dies to be sized analytically on the first iteration
- Owned a work cell and wrote detailed work instructions with technicians covering cable assembly, joint placement, orbital cutting, and welding, replacing tribal knowledge with a repeatable reference used on the production floor

Raytheon Missiles & Defense Andover, MA
Operations Engineering Co-op *Jan 2024 – Sep 2024*

- Prototyped a composite fixture using **SolidWorks** and PCB Gerber files to shield heat-sensitive filters from 93 °C while enabling reflow of adjacent components, **increasing first-pass yield** and removing filter damage as a scrap driver on the line
- Ran a nine-configuration solder stencil study varying placement and volume to provide off-gas escape channels, reducing voiding and **removing a rework** step from the process
- **Co-owned a work cell** and developed thermal reflow profiles by refining oven parameters through thermocouple data, operator input, and X-ray inspection to IPC-J-STD-001 Class 3, **reducing rework time across 15 programs**

Projects

Underwater Acoustic Release Mechanism – Capstone Project Boston, MA
Mechanical Design Lead – Release System *Jun 2026 – Present*

- **Own the mechanical design** of a low-cost underwater acoustic release mechanism for underwater robot recovery; designing a rotary lever arm mechanism and using first-principles force analysis to evaluate and rule out initial design concepts
- Performed **structural analysis** of the release shaft under combined axial compression and torsion, and selected O-ring seal geometry using Parker's O-Ring Handbook to keep the rotating, pressure-sealed shaft interface leak-free at 300 m depth
- Validating hand-calculated stress predictions on housing and latch components in **Ansys FEA** under combined pressure and mechanical loading

AerospaceNU – Liquid Rocket Engine Boston, MA
Propulsion - Feed System Team *Sep 2023 – Present*

- Designing and validating an actively controlled nitrogen pressurant subsystem, integrating a servo-actuated ball valve and torque coupling with closed-loop pressure regulation based on downstream pressure feedback
- Supported setup and execution of **three successful static-fire tests**, integrating propulsion, electrical, and instrumentation systems while configuring test hardware and data acquisition equipment

Technical Skills

Software: Siemens NX | SolidWorks | Ansys Mechanical | Arduino | Teamcenter | SolidWorks PDM | Odoo | Confluence | Jira

Design: Structural Analysis | DFM | GD&T | FEA (Ansys – developing)

Fabrication: Manual G-Code | CNC Milling | Power Tools | TIG Welding | 3D Printing | Soldering | Water Jetting | Laser Cutting